Hai-Xia Ma — Publication List

First-author

[1] **H.-X. Ma**, T. T. Takeuchi, S. Cooray, and Y. Zhu. *sOPTICS: A Modified Density-Based Algorithm for Identifying Galaxy Groups/Clusters and Brightest Cluster Galaxies*. In: *Monthly Notices of the Royal Astronomical Society* 537.2 (Feb. 2025), pp. 1504–1517. ISSN: 0035-8711. DOI: 10.1093/mnras/staf115.

[2] Y. Zhu, **H.-X. Ma (co-first author)**, X.-B. Dong, Y. Huang, T. Mistele, B. Peng, Q. Long, T. Wang, L. Chang, and X. Jin. *How Close Dark Matter Haloes and MOND Are to Each Other: Three-Dimensional Tests Based on Gaia DR2*. In: *Monthly Notices of the Royal Astronomical Society* 519 (Mar. 2023), pp. 4479–4498. ISSN: 0035-8711. DOI: 10.1093/mnras/stac3483.

Co-author

[3] T. T. Takeuchi, **H.-X. Ma**, and Y. Ogane. *Event-driven formulation of cosmic structure formation via Morse-Cerf theory*. In: *Phys. Rev. D* (Oct. 2026). DOI: 10.1103/k2zn-96wd.

[4] Y. Zhu, M. J. Rieke, L. C. Ho, Y. Sun, G. H. Rieke, F. Yuan, T. J. L. C. Bakx, G. D. Becker, J. Yang, E. Bañados, M. Bischetti, C. Cain, X. Fan, Y. Fudamoto, S. Hashemi, R. Ikeda, Z. Ji, X. Jin, W. Liu, Y. Liu, J. Lyu, **H.-X. Ma**, T. T. Takeuchi, H. Umehata, F. Wang, and W. L. Tee. *A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars*. In: *ApJ* 1000.2 (Mar. 2026), p. 312. ISSN: 0004-637X. DOI: 10.3847/1538-4357/ae4ed1.

[5] Y. Zhu, E. Egami, X. Fan, F. Sun, G. D. Becker, C. Cain, H. Chen, A.-C. Eilers, Y. Fudamoto, J. M. Helton, X. Jin, M. Pudoka, A. J. Bunker, Z. Cai, J. B. Champagne, Z. Ji, X. Lin, W. Liu, **H.-X. Ma**, Z. Ma, R. Maiolino, G. H. Rieke, M. J. Rieke, P. Rinaldi, Y. Sun, W. L. Tee, F. Wang, J. Yang, M. Yue, and J. Zhang. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*. In: *The Astrophysical Journal* 995.1 (Dec. 2025), p. L5. ISSN: 0004-637X. DOI: 10.3847/2041-8213/ae1f8e.

[6] X. Dong, Y. Zhu, M. Rieke, G. Rieke, X. Li, P. Behroozi, **H.-X. Ma**, R. Meng, Z. Mao, and Z. Sun. *A Promise for the JWST Era: Massive Black Holes Directly Collapsed from Wave Dark Matter Haloes, and Star Formation in and around Their Accretion Flows*. Aug. 2025. DOI: 10.48550/arXiv.2508.09258. arXiv: 2508.09258 [astro-ph].

[7] Y. Zhu, T. J. L. C. Bakx, R. Ikeda, H. Umehata, G. D. Becker, C. Cain, J. B. Champagne, X. Fan, Y. Fudamoto, X. Jin, **H.-X. Ma**, Y. Sun, T. T. Takeuchi, and W. L. Tee. *Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at $z = 5.63$*. In: *Res. Notes AAS* 8.11 (Nov. 2024), p. 284. ISSN: 2515-5172. DOI: 10.3847/2515-5172/ad91ad.

[8] Y. Zhu, G. D. Becker, H. M. Christenson, A. D'Aloisio, S. E. I. Bosman, T. Bakx, V. D'Odorico, M. Bischetti, C. Cain, F. B. Davies, R. L. Davies, A.-C. Eilers, X. Fan, P. Gaikwad, M. G. Haehnelt, L. C. Keating, G. Kulkarni, S. Lai, **H.-X. Ma**, A. Mesinger, Y. Qin, S. Satyavolu, T. T. Takeuchi, H. Umehata, and J. Yang. *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*. In: *The Astrophysical Journal* 955 (Oct. 2023), p. 115. ISSN: 0004-637X. DOI: 10.3847/1538-4357/aceef4.

[9] S. Cooray, T. T. Takeuchi, D. Kashino, S. A. Yoshida, **H.-X. Ma**, and K. T. Kono. *Characterizing and Understanding Galaxies with Two Parameters*. In: *Monthly Notices of the Royal Astronomical Society* 524 (Oct. 2023), pp. 4976–4995. ISSN: 0035-8711. DOI: 10.1093/mnras/stad2129.